

Deep Learning-Based Detection of Abnormal Axial Cross-Sectional Planes in Fetal Echocardiography

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Abstract—Early diagnosis of congenital heart diseases (CHDs) via fetal echocardiogram ultrasonography (US) is critical for minimizing infant mortality and enabling timely medical interventions. In this research, we proposed a deep learnin method that examines many fetal heart cross-sections at once, namely the four-chamber-view (4CV) and the three-vessel-and-trachea-view (3VTV). Three deep learning models: DenseNet169, MobileNetV2, and Xception — were trained using our dataset, which included 1,781 pictures, to categorize fetal heart images into three classifiers: normal, 4CV abnormal, and 3VTV abnormal. High classification performance was shown by the experimental findings, with individual models obtaining accuracies ranging from 91% to 94%. Our method presents a more thorough evaluation of embryonic heart problems by utilizing various ultrasound planes, which may enhance early detection, direct clinical judgments, and ultimately improve mother and fetal health outcomes.

Keywords—Congenital Heart Disease, Deep Learning, Ultrasound Imaging, Convolutional Neural Networks.

I. INTRODUCTION

Congenital heart diseases (CHDs) represent the most prevalent category of birth defects, impacting roughly 1% of live births globally. Early prenatal diagnosis of CHDs through fetal heart screening allows for timely intervention and improved postnatal management. Ultrasound imaging plays a crucial role in fetal heart screening, enabling the visualization of various cross-sectional planes of the fetal heart. However, the accurate identification of abnormal planes often requires significant expertise and might be time-consuming and subject to inter-observer variation.

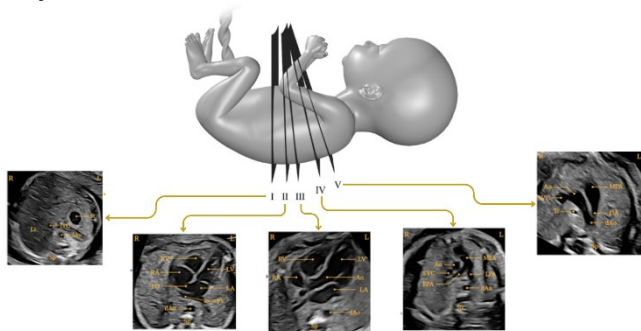


Fig.1. Five plans: (I) Most caudal plane, (II) Four-chamber view, (III) Left ventricular outflow-tract view (LVOT), (IV) Right ventricular outflow-tract view (RVOT), (V) Three-vessel-and-trachea view [1].

Referring to the International Society of Ultrasound in Obstetrics and Gynecology (ISUOG), it is an academic institution that promotes good clinical practice, excellent instruction, and research in the field of diagnostic imaging for

women's health; and fetal ultrasound experts on cardiac structure, we determined that ultrasound cross-sections can be divided during fetal echocardiography into five post-muscular cross-sections as shown in Figure 1.

After researching the structure and cases of fetal malformations, we realized that detecting fetal heart abnormalities will often be identified based on two main cross-sections: the four-chamber-view (4CV) and the three-vessel-and-trachea-view (3VTV). This abnormality can be detected by abnormalities of the structural components of the heart or also by the color of the flow during the measurement of blood flow using the Doppler ultrasound technique. It can be seen that different types of congenital malformations can appear from different positions according to different angles of the image or ultrasound fetal video. Because of this difficulty, we want to aim at identifying these cross-sections in the research, not just the cross-sections but also being able to determine which cross-sections have signs of abnormalities. Here, we use three main classes for the research model: two classes with cross-sections showing signs of abnormalities, including 4CV and 3VTV, and the remaining class is the images containing cross-sections, but no signs of abnormalities are detected. In addition, using deep learning to apply to medicine is also the main research we want to aim for. The use of machine learning in medicine is also one of the fields in which the scientific community aims for more open development. Deep learning, particularly Convolutional Neural Networks (CNNs), has demonstrated extraordinary performance in numerous medical image processing applications [2]-[6].

There is growing interest in creating automated methods for evaluating fetal cardiac ultrasound images. There have been outstanding studies related to related to fetal echocardiography images and videos. Detect structural heart components in both normal and abnormal congenitally malformed ultrasound videos by SONO [7] or YOLOv7 [8]. These models focus on plane-based fetal heart detection and the input data depends on the skill level of the examiner. However, non-experts continue to struggle with identifying the cardiac accessory structures and accurately describing the scanning planes. Detect congenital heart disease types through four labels such as ASD, AVSD, VSD, and one normal using a stacked 8 CNN model [9], or up to 7 labels for CHDs types using the DenseNet201 structure [10]. As can be seen, the studies emphasize the complexity of the model and data. Most of the deep learning-based cardiac anomaly detection is still problematic due to CHD are comparatively rare, besides many factors affecting ultrasound images, such as image quality, resolution, graininess, noise, etc. [11] pathology of the pregnant woman's development [12] prepare a complete

training dataset a problematic challenge. The value of detecting early malformations in the first trimester of pregnancy is also significant [13], [14]. Specialists are able to select the best course of treatment for their patients by precisely identifying and detecting the location. Synthesizing the above issues, we have considered another aspect to identify fetal echocardiography abnormalities on two 4CV and 3VTV cross-sections based on the diversity of ultrasound images, gestational weeks, and other angles.

This paper proposes using different CNN models, which are fine-tuned for the specific task for experimentation and analysis on our dataset. We use deep learning to train the model and test which model to use on the dataset many times. With a small and medium dataset, fetal echocardiography images have many complex features and are affected by many different factors; we selected three models to test MobileNetV2, Xception, and DenseNet169 with an accuracy rate of approximately 91-95 percent.

In summary, this research makes the following contributions:

- Due to the scarcity of fetal echocardiography data and its unavailability for this research topic, we create a new dataset for fetal echocardiography due to lack of data, including normal and abnormal images (4CV, 3VTV), annotated by experts.
- The experiment is based on three different CNNs models to classify data into three categories (normal, 4CV abnormal, 3VTV abnormal). The results yielded performance ranging from 91 to 95%.

II. DATA PREPARATION

The foundation of our study relies on the quality and preparation of our dataset. This section outlines the detailed steps involved in preparing the ultrasound images for training and evaluating our proposed ensemble model.

A. Image Acquisition

The dataset comprises a collection of fetal heart ultrasound images acquired from various clinical settings. These images were obtained using standard ultrasound equipment following established clinical protocols for fetal echocardiography. The dataset includes images from fetuses across a range of gestational ages (shown in TABLE I), typically within the second and third trimesters, when detailed cardiac anatomy can be visualized effectively. The acquisition process adhered to guidelines ensuring the capture of key cross-sectional planes essential for cardiac evaluation. These planes primarily include the four-chamber view (4CV), which provides a comprehensive overview of the heart's chambers and valves, and the three-vessel trachea view (3VTV), crucial for assessing the outflow tracts and great arteries. The dataset encompasses both normal and abnormal cases, representing a diverse spectrum of fetal cardiac conditions. 4CV represents CHDs involving the four chambers of the heart such as HLHS, VSD, ASD, AVSD, TOF, Coarctation of aorta, Atrial septal aneurysm, ... while 3VTV will include more complex CHDs involving the vessels and lungs such as Persistent left superior vena cava (PLSVC), 2 vessels dilated, 4 vessels, Dilated coronary sinus, Double aortic arch, Dilated pulmonary ...

We collected and extracted ultrasound images with the smallest being 150x150 pixels and the largest being 2000x1500 pixels from publicly available datasets and images

collected according to standard medical protocols from popular ultrasound machines of Phillip and GE healthcare. This will result in images that are not uniform in size and quality. However, such image diversity allows for improved generalization of the models we tested, resulting in performance over a wider range of information.

B. Annotation

Accurate and reliable annotation is paramount for supervised learning tasks. In this study, the ultrasound images were meticulously annotated by experienced clinical experts specializing in fetal echocardiography. These experts carefully reviewed each image and assigned labels indicating whether the visualized cross-sectional plane was normal or exhibited characteristics indicative of an abnormality. Specifically, the annotation process involved identifying key anatomical landmarks within the 4CV and 3VTV images to determine the presence or absence of structural defects. To ensure the consistency and reliability of the annotations, a standardized annotation protocol was followed, and in cases of ambiguity, consensus was reached through joint review by multiple experts. This rigorous annotation process ensures the ground truth labels used for training and evaluation are of high quality.

TABLE I. STATISTICAL TABLE OF FETAL ECHOCARDIOGRAM IMAGES BY WEEK OF AGE.

Trimester	Gestational Week	Cardiac Size (cm)	Images
First trimester	01 – 12	< 1	0
Second trimester	13 – 20	~ 1 - 2	37
	21 – 26	~ 2.5	358
Third trimester	27 – 34	~ 2.5 and more clearly	145
	35 – 40	~ 3 – 4	23

C. Dataset Augmentation

To boost the generalizability and robustness of our CNN models, we employ bilinear interpolation to rotate the entire original data by ± 5 degrees. This method creates a smoother image and also reduces aliasing with reasonable processing speed. This method is implemented with the following procedure:

1. *Determine Target Coordinates:* For each pixel (x', y') in the rotated output image, find its corresponding location (x, y) in the original image. The inverse rotation formula is typically used:

$$x = x' \cos(\theta) + y' \sin(\theta)$$

$$y = -x' \sin(\theta) + y' \cos(\theta), (\theta \text{ is the rotation angle})$$

2. *Find Four Neighboring Points:* Find the four nearest neighboring pixels in the original image: (x_1, y_1) , (x_2, y_1) , (x_1, y_2) and (x_2, y_2) , such that $x_1 \leq x < x_2$ and $y_1 \leq y < y_2$.

3. *Interpolate Twice:*

- Horizontal Interpolation (x): Estimate the pixel value at (x, y_1) by linearly interpolating between (x_1, y_1) and (x_2, y_1) . Similarly, estimate the value at (x, y_2) from (x_1, y_2) and (x_2, y_2) .

- Vertical Interpolation (y): Use the two values estimated in step 1 (at (x, y_1) and (x, y_2)) to linearly interpolate one more time, yielding the final pixel value for the target pixel (x', y') in the rotated image.

Fig. 2 simulates the architecture implementing the bilinear interpolation method of the study.

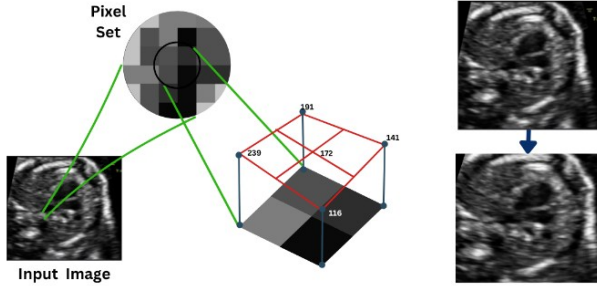


Fig. 2. Bilinear interpolation method architecture

D. Image Filtering and Preprocessing

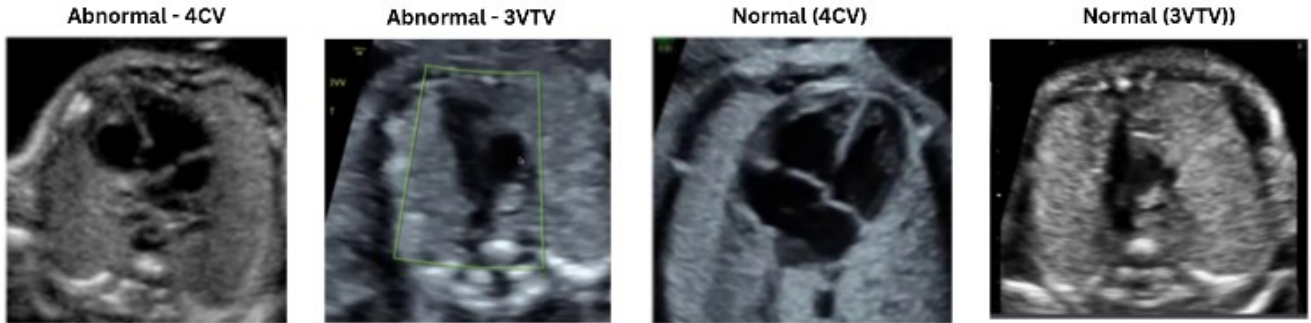


Fig. 3. Three classifiers of Fetal echocardiogram images in dataset: 2 abnormal classes in the 4CV and 3VTV and one normal class.

Prior to feeding the ultrasound images into the CNN models, several preprocessing steps were performed to improve image quality and focus the models on the relevant information. The entire image will be filtered to match the appropriate images, removing excess background to focus on the points that need to be identified. Finally, the images will be resized to 224x224. Finally, our dataset contains 1781 photos, with 577, 475, and 729 in the normal, 4CV, and 3VTV classes, respectively. The image representing the data set is shown in Fig. 3.

These detailed data preparation steps ensure that the CNN models are trained on a high-quality, well-annotated, and diverse dataset, which is crucial for achieving robust and accurate detection of abnormal fetal heart cross-sectional planes.

III. METHODOLOGY

This section elaborates on the proposed approach for detecting abnormal fetal heart cross-sectional planes, which involves three different Convolutional Neural Networks (CNNs). The overall workflow includes experimentation, analysis and evaluation of results, discussion of individual CNN models.

A. CNN Model Architecture

In this study, we describe in detail three distinct CNN architectures, each with unique characteristics suitable for image classification tasks:

1. DenseNet169: This architecture features dense connections where each layer is linked to all preceding layers. This dense connectivity promotes feature reuse, strengthens gradient propagation, and can lead to improved performance, especially in deep networks, by encouraging the network to learn redundant features.
2. MobileNetV2: Known for its efficient architecture designed for mobile and resource-constrained applications, MobileNetV2 utilizes inverted residual blocks and linear bottlenecks to minimize computational cost while maintaining accuracy. Its depthwise separable convolutions are key to its efficiency.
3. Xception: An architecture inspired by Inception, Xception replaces standard convolutional layers with depthwise separable convolutions arranged in a specific structure. This design aims to improve efficiency and performance by explicitly modeling channel-wise and spatial convolutions separately.

Each of these pre-trained CNN models was initialized with weights learned from training on the ImageNet dataset. This pre-training allows the models to start with a rich set of general image features, which can then be fine-tuned for the specific task of fetal heart ultrasound image classification.

B. Training and Evaluation of Individual CNN Models

The model was implemented using PyTorch1 and trained on a single NVIDIA RTX 4090 GPU, with 24GB RAM. The operating system was based on Ubuntu, with Python 3 as the primary programming language. All image data is resized to 224x224 pixels before being supplied into the training network. The training parameters set for the structure of the base learner models is shown in TABLE :

TABLE II. THE HYPERPARAMETER VALUES USED TO TRAIN THE BASE CNN CLASSIFIERS

Hyperparameter	Value
Input Shape	224x224
Optimizer	RMSProp
Learning rate	2.00E – 0.5
Batch size	32
Dropout rate	0.2
Loss function	Categorical Cross Entropy

After training was complete, the trained CNN model was used to predict the class labels (normal or abnormal) for the images in the dataset. The equations will produce the same matrix with positive and negative binary classification classes. The numbers of True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) will be presented. To demonstrate the performance of each individual base learner will be compared with four popular evaluation criteria:

- Accuracy: The percentage of correctly classified images.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}$$

- Precision: The proportion of accurately anticipated abnormal instances to the total number of predicted abnormal cases.

$$PRE = \frac{TP}{TP + FP}$$

- Recall (Sensitivity): The proportion of accurately anticipated abnormal cases to the total number of observed abnormal cases.

$$REC = \frac{TP}{TP + FN}$$

- F1-Score: The harmonic mean of precision and recall, which provides a balanced measure of performance.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

This entire training and evaluation process was independently carried out for each of the three CNN architectures: MobileNetV2, Xception, and DenseNet169. By following this methodology, we were able to assess the individual capabilities of each model in detecting abnormal fetal heart cross-sectional planes without the influence of any ensemble fusion technique. The results of this evaluation would provide insights into the strengths and weaknesses of each architecture for this specific medical image analysis task.

IV. RESULT

The results from TABLE III indicate that all three CNN models achieved encouraging performance in classifying fetal heart cross-sectional planes. However, there are notable differences in performance among the architectures.

TABLE III. PERFORMANCE OF INDEPENDENT CNN MODELS ON THE DATASET

CNN	Class	PRE	REC	F1	Sup	ACC
Dense Net 169	Normal	93.28	95.69	94.47	116	94.94
	4CV	95.74	94.74	95.24	95	
	3VTV	94.59	95.89	95.24	145	
	AVG	94.96	94.94	94.95	356	
Mobile NetV2	Normal	92.86	89.66	91.23	116	93.26
	4CV	95.74	94.74	95.24	95	
	3VTV	92.00	95.17	93.56	145	
	AVG	93.28	93.26	93.25	356	

Xception	Normal	93.33	84.48	88.69	116	91.57
	4CV	97.73	90.53	93.99	95	
	3VTV	87.12	97.93	92.21	145	
	AVG	91.97	91.57	91.54	356	

As we mentioned before, the dataset has a variety of sizes, gestational weeks, images from different machine lines, although there is image synchronization before testing the models, there will still be performance differences when some images are too small or low gestational weeks bring high noise. DenseNet169 with its special architecture has brought higher rates but because of the above problems, there will still be loss as well as confusion between layers. We have adjusted the dense layers with 512 units, which is suitable for the model and gives the most stable performance results.

DenseNet169 demonstrated the highest performance across all metrics, achieving an accuracy of nearly 95.0%. Besides, the remaining indicators when we run the average of the 3 classes also bring high rates, 94.96% 94.94% and 94.95% for Precision, Recall and F1-score respectively. This shows that DenseNet169 provides a consistent and reasonable prediction rate and performance. In particular, DenseNet169 achieves a good balance between Precision and Recall for both 4CV and 3VTV classes (F1-Score is 95.24%). This shows that DenseNet169 is capable of accurately recognizing both types of abnormal cross-sections without a significant bias between missing the actual case (Recall) and mispredicting the normal case (Precision). The dense connectivity of the DenseNet architecture, which promotes feature reuse and efficient gradient flow, likely contributed to its superior ability to recognize complex patterns in the fetal ultrasound images.

MobileNetV2, designed for computational efficiency, still achieved a respectable average accuracy of 93.26%, a precision of 93.53%, a recall of 93.19% and an F1-score of 92.7%, lower than DenseNet169. The performance of VGG16 varies between classes. The F1-Score for class 4CV is the highest (95.24%), however, the lack of Precision and Recall information for this class makes it difficult to fully evaluate. For class Normal, Recall is lower than Precision, suggesting that the model may miss some normal cases. The performance for class 3VTV is quite good with an F1-Score of 93.56%. While not achieving the highest performance, MobileNetV2 demonstrates a good trade-off between accuracy and efficiency, making it a potentially suitable option for resource-constrained applications.

Xception achieved the lowest overall accuracy of the three models, rates up to 91.57%, 91.97%, 91.57% and 91.54% for accuracy, precision, F1-score and accuracy respectively. It is worth noting the large difference between Precision and Recall for the 4CV and 3VTV classes. For the 4CV class, Precision is very high (97.73%) but Recall is lower (90.53%), indicating that the model is very confident in predicting an image as 4CV but may miss some actual 4CV instances. Conversely, for the 3VTV class, Recall is very high (97.93%) but Precision is lower (87.12%), indicating that the model tends to predict more instances as 3VTV than actually. Xception's architecture, based on depthwise separable convolutions, may have effectively learned the spatial and channel-wise correlations in the data, leading to accurate classification. The balance between performance and the number of parameters in Xception could be an advantage in practical applications.

The differences in performance can be attributed to the variations in the architectures and how these models learn features from the data. Overall, DenseNet169 showed the best and most consistent performance in classifying fetal heart cross-sections in this dataset. This may be due to the fact that the DenseNet architecture encourages feature reuse and improves gradient information flow, helping the model learn rich representations and better discriminate between classes. DenseNet169 with their more complex designs and capacity to learn richer feature representations, outperforms MobileNetV2 and Xception. MobileNetV2 and Xception also achieve significant performance, but have specific weaknesses. MobileNetV2 may have difficulty classifying the Normal class, while Xception shows a bias between Precision and Recall for the Abnormal classes.

The training performance progress is shown in Figure 4 and Figure 5 displays the confusion matrix utilizing the three proposed CNN methods. In Figure 5, the confusion matrix is shown for Normal (0), 4CV (1) and 3VTV (2). By comparing the values on the main diagonal and the values off the main diagonal between the three matrices, we can get an overall assessment that the DenseNet169 method classifies the 4CV class (137 correct) and 3VTV class (111 correct), with fewer

confusion errors than the Normal class. MobileNetV2 is quite similar to DenseNet169 but seems to be slightly worse at the 3VTV class (104 vs 111). Xception has the highest number of correct 4CV predictions (142), but has a lot of confusion errors in the 3VTV class (16 3VTV samples were mistaken for 4CV, and only 98 3VTV samples were predicted correctly). In summary, we evaluate the DenseNet169 method as stable and providing the highest performance.

The high performance of all three models suggests the significant potential of deep learning in assisting medical professionals with fetal heart screening. These models could be integrated into diagnostic support systems to provide objective information and help improve the accuracy and effectiveness of the screening process.

This study focuses on three specific CNN architectures. Exploring additional architectures or variations within could potentially yield further improvements. Additionally, the performance of the models is likely dependent on the quality and diversity of the training dataset. Expanding the dataset with more cases and different types of abnormalities could enhance the generalizability of the models.

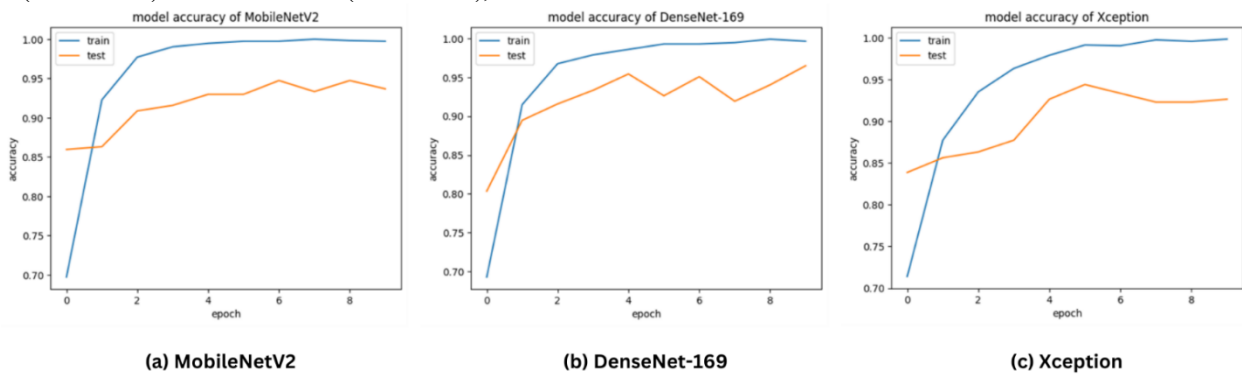


Fig.4. Training process of the three CNNs: MobileNetV2, DenseNet-169, and Xception

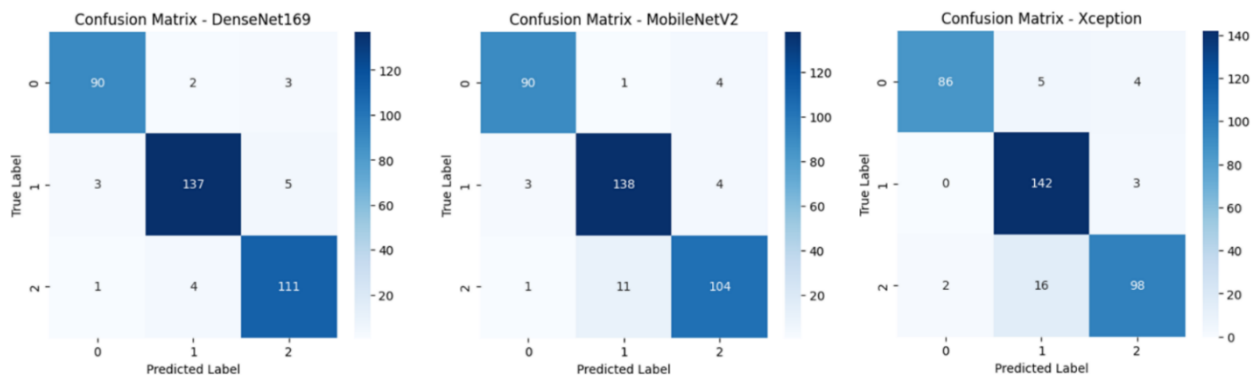


Fig. 5. Confusion matrix encompassing three CNN model with three classifiers: Normal (0), 4CV (1), 3VTV (2).

V. CONCLUSION

The experimental results demonstrate that DenseNet169, MobileNetV2, and Xception are all effective methods for detecting abnormal fetal heart cross-sectional planes. DenseNet169 achieved the highest performance, followed by MobileNetV2 and then Xception, highlighting the trade-off between performance and computational efficiency. These findings provide valuable insights for selecting appropriate models for practical fetal heart screening applications.

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